Joint Distributions $\Longrightarrow$ Marginal & Conditional Densities

X, Y two random variables

$$P(x,y) = P(X=x, Y=y)$$

$$P_X(x) = \sum_j P(X=x, Y=y_j)$$

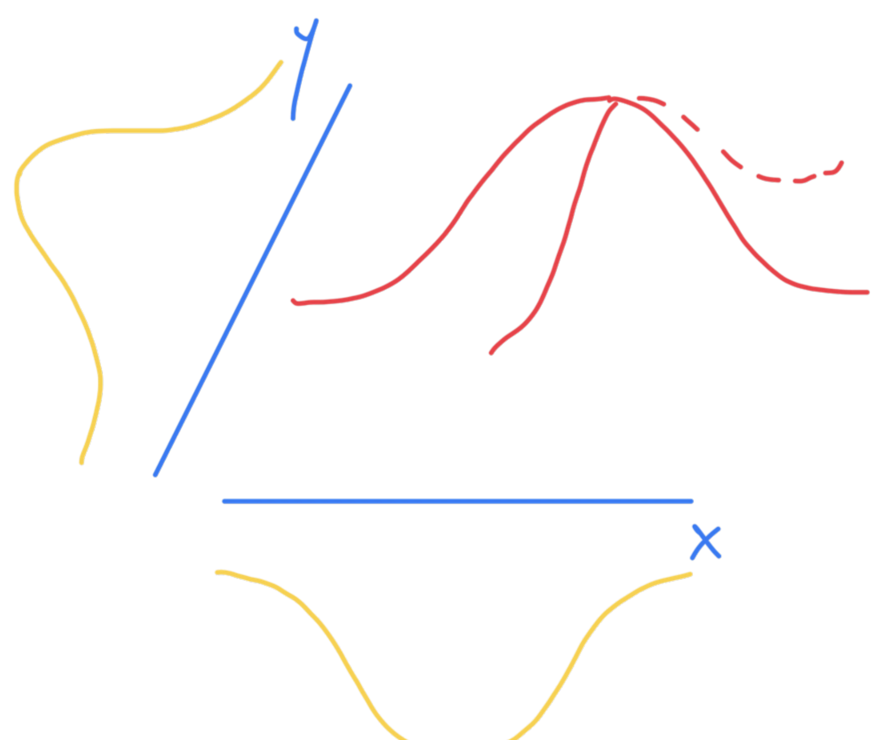
$$P((x,y)\in A) = \int_A \overbrace{f(x,y)}^{\text{joint PDF}}\, dx\, dy$$

$\Rightarrow$ Marginal Density

$$f_X(x) = \int_{-\infty}^{\infty} f(x,y)\, dy$$

$$f_Y(y) = \int_{-\infty}^{\infty} f(x,y)\, dx$$

$$\underbrace{X, Y \sim N}_{f(x,y)}$$



$$f(x,y) = \frac{1}{\sqrt{2\pi}}\, e^{-\frac{(x^2+y^2)}{2}}$$

$$P(X=x \mid Y=y) = \frac{P(x,y)}{P_Y(y)}$$

$$f_{Y|X}(y|x) = \frac{f(x,y)}{f_X(x)}$$